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Logistic Regression: building classification model

Logistic regression is a machine learning classification model with similar as Linear Regression

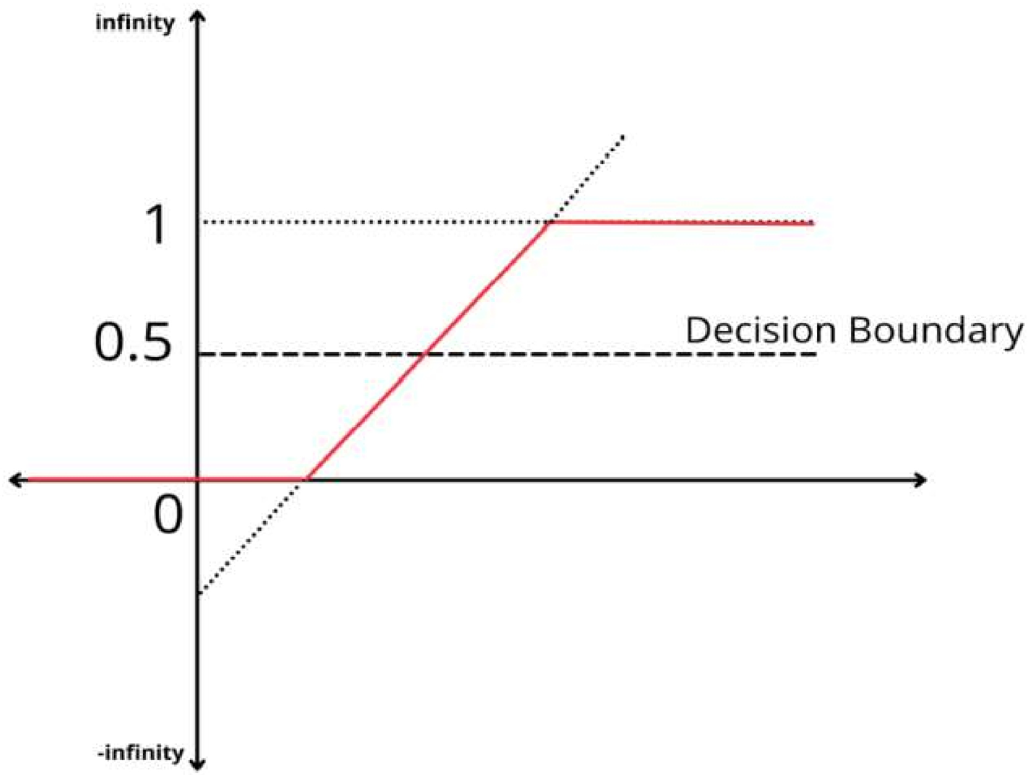
but it's not used to predict a continuous outcome. -Not for Regression.

Instead, it is a statistical BINARY classification model.

Binary classes defined as success and failure: -

0 (Failure) and 1 (Success), Yes / No ,Selected / Rejected ,True/ False

Decision boundary:



The S-shaped curve is a sigmoid curve. Logistic regression function is also called sigmoid function.

p is a non-linear function, say, an exponential function.

Sigmoid Function: $$\boldsymbol{p = \frac{1}{1 + e^{-y}}}$$

For -∞

$$p = \frac{1}{1 + e^{-(-\infty)}}$$

$$p = \frac{1}{\infty}$$

Anything divided by ∞ is 0, Hence for -∞ it is **p = 0**

$$p = \frac{1}{1 + e^{-y}}$$

For ∞

$$p = \frac{1}{1 + e^{-(+\infty)}}$$

$$p = \frac{1}{1 + 0}$$

Hence for ∞ it **is p = 1**

logistic regression formula

Sigmoid function:-

Mathematical function having a characteristic "S"shaped curve or sigmoid curve.

$$p = \frac{1}{1 + e^{-y}}$$

Transformation of the sigmoid function

$$p(1 + e^{-y}) = 1$$

$$p + pe^{-y} = 1$$

$$pe^{-y} = 1 - p$$

$$e^{-y} = \frac{1 - p}{p}$$

$$log(e^{-y}) = log\left(\frac{1 - p}{p}\right)$$

$$-y = ln\left(\frac{1 - p}{p}\right)$$

$$y = ln\left(\frac{p}{1 - p}\right)$$

Hence we have converted the sigmoid function such that now it is expressed as providing the value of Y that is the target variable

$$y = ln\left(\frac{p}{1 - p}\right)$$

Here (P/ (1-P)) is called "odds ratio".

Hence Y can be defined as the log of the odds ratio.

Y is also called as Log Odd Function or Logistic Function or Log it Function

Applying Linear Equation Formula

$$exp\left(log\left(\frac{p}{1-p}\right)\right) = exp\left(a + b_1 x_1\right)$$

$$\Rightarrow \frac{p}{1-p} = exp\left(a + b_1 x_1\right) \quad [Since\ exp(logx) = x]$$

Solving the above equation for p, we get

$$\mathrm{p} = exp\left(a + b_1 x_1\right) - \mathrm{p}\ exp\left(a + b_1 x_1\right)$$

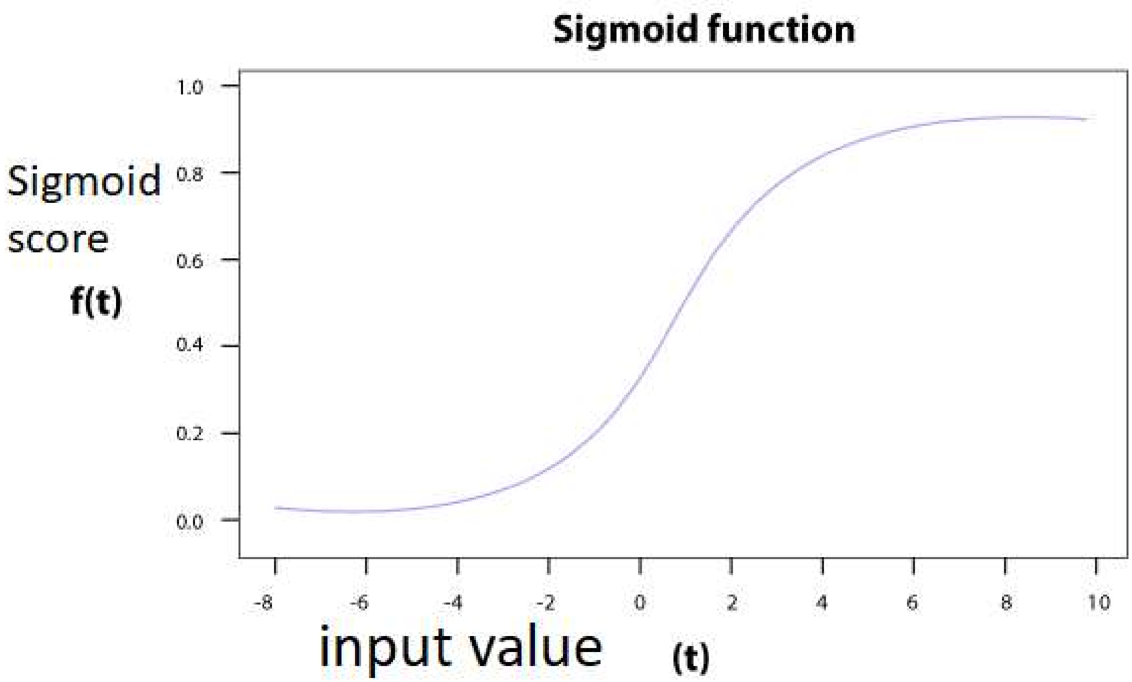
$$p = \frac{exp(a+b_1 x_1)}{1+exp(a+b_1 x_1)}$$

Finally, we will get the value of probability p in the range of 0 to 1.

Once we calculate the probabilities, we will plot them to get an s-curve. Now,
We just keep rotation the log(odds) line and projecting the data points onto it and then transforming it to probabilities and calculating the log-likelihood. We will repeat this process until we maximize the log-likelihood.

Keeping Threshold =0.5
Above 0.5 output belongs to class 1

Below 0.5   output belongs to class 0




```
from sklearn.linear_model import LogisticRegression
lr=LogisticRegression()
lr.fit(X_train1,Y_train)

Y_pred_lr_train=lr.predict(X_train1)
Y_pred_lr_test=lr.predict(X_test1)

print("LOGISTIC REGRESSION TRAINING ACCURACY ",accuracy_score(Y_train,Y_pred_lr_train))

print("######"*20)

print("LOGISTIC REGRESSION TESTING ACCURACY ",accuracy_score(Y_test,Y_pred_lr_test))
```

MATHEMATICS of Logistic Regression:-

Maximum Likelihood Estimation

logistic regression formula
Sigmoid function:-
Mathematical function having a characteristic "S"shaped curve or sigmoid curve.

$$p = \frac{1}{1 + e^{-y}}$$

The logistic regression function converts the values of **logits** also called **log-odds** that range from **–∞ to +∞** to a range between **0 and 1**.

Transformation of the sigmoid function

$$p(1 + e^{-y}) = 1$$

$$p + pe^{-y} = 1$$

$$pe^{-y} = 1 - p$$

$$e^{-y} = \frac{1 - p}{p}$$

$$log(e^{-y}) = log\left(\frac{1 - p}{p}\right)$$

$$-y = ln\left(\frac{1 - p}{p}\right)$$

$$y = ln\left(\frac{p}{1 - p}\right)$$

Odds is defined as the ratio of the probability of occurrence of a particular event to the probability of the event not occurring.

$$Odds = \frac{P}{1 - P}$$

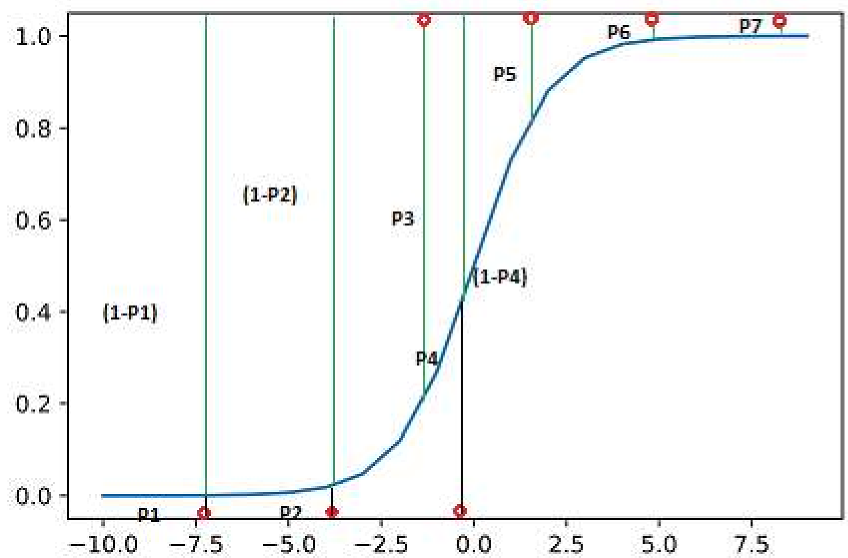
$$\ln(Odds) = \ln\left(\frac{P}{1-P}\right) = y$$

$$\ln\left(\frac{P}{1-P}\right) = \beta 0 + \beta 1x$$

Logistic Regression predicts output variable as 0 or 1, we need to find the best fit sigmoid curve.

Now a cost function tells how close the values are from actual. So here we need a cost function which maximizes the likelihood of getting desired output values. Such a cost function is called as **Maximum Likelihood Estimation (MLE)** function.

The cost function should be such that it maximizes the probability of predicted values being close to the actual ones.



Sigmoid curve and probabilities

From the above figure, we can see the points being classified as 0 or 1 and the respective probabilities associated with them. There are 7 points and seven associated probabilities P1 to P7.

For points to be 0, we need the probabilities P1, P2 and P4 to be as minimum as possible and for points to be 1, we need the probabilities P3, P5, P6 and P7 to be as high as possible, for correct classification. We can also say that (1-P1), (1-P2), P3, (1-P4), P5, P6 and P7 should be as high as possible.

The joint probability is nothing but the product of probabilities. So the product:[**(1-P1)*(1-P2)* P3*(1-P4)*P5*P6*P7]** should be maximum.

This joint probability function is our cost function or **the Maximum Likelihood Estimation (MLE)** function.

This joint probability function is nothing but our cost function which should be maximized in order to get a best fit sigmoid curve. Or we can say predicted values to be close to the actual values.

Now coming to our cost function, let **J(z)** be a function of **z** such that

J(z) = P(Y ; z) = P(Y1, Y2 ... Yn ; z)

The assumption here is that all Y are independent.

$$J(z) = \prod_{i=1}^{n} P(Yi\ ;\ z)$$

Cost function is product of all probabilities P(Yi)

Taking natural log on both sides :

$\ln(J(z)) = \ln\left(\prod_{i=1}^{n} P(Yi\,; z)\right)$

Taking log of both sides

Since log of product becomes summation :

$\ln(J(z)) = \sum_{i=1}^{n} \ln\left(P(Yi\,; z)\right)$

log of product is summation

J(z) can also be written as L(z|Yi) (L for Likelihood). For a given value of **z** and observed sample Yi , this function gives the probability of observing the sample values. So if Yi=1 the expression becomes **z** and if Yi is 0 the expression becomes **1-z**:

$\ln(J(z)) = L(z \mid Yi) = \sum_{i=1}^{n} \ln\left(z^{Yi} * (1-z)^{1-Yi}\right)$

Solving this equation further we get :

$\ln(J(z)) = L(z \mid Yi) = \sum_{i=1}^{n} Yi * ln(z) + (1 - Yi) * ln(1 - z)$

Differentiating this equation with respect to **z** and setting the derivative to zero, we calculate the maxima using closed form solution:

$$\frac{d}{dP}\left(\sum_{i=1}^{n} Yi * ln(z) + (1 - Yi) * ln(1 - z)\right) = 0$$

Solving further, this equation becomes:

$$\frac{z}{1-z} = \sum_{i=1}^{n} \frac{Yi}{1-Yi}$$

The right side term represents the ratio of number of 1s to number of 0s. Thus the function achieves a maximum at :-

$$z = \sum_{i=1}^{n} \frac{Yi}{n}$$

Binary Classification

The following are a few binary classification applications, where the 0 and 1 columns are two possible classes for each observation:

Application	Observation	0	1
Medical Diagnosis	Patient	Healthy	Diseased
Email Analysis	Email	Not Spam	Spam
Financial Data Analysis	Transaction	Not Fraud	Fraud
Marketing	Website visitor	Won't Buy	Will Buy
Image Classification	Image	Hotdog	Not Hotdog
Employee Churn	HRD-Employee	Continue	Churn
Tumor	Website visitor	Benign	Malignant

CLASSIFICATION MODEL EVALUATION

CONFUSION MATRIX

Actual		
	True Positive(TP) #Reality: A wolf threatened # Shepherd said: "Wolf" #Outcome:Shepherd is a Hero.	False Positive(FP) #Reality : No wolf threatend. #Shepherd Said:"Wolf" #Outcome:Villagers are angry at shepherd for waking them up.
	False Negative(FN) # Reality: A wolf threatend. #Shepherd said:"No Wolf" #Outcome:The wolf ate all sheep.	True Negative(TN) #Reality: No wolf threatend #Shepherd said:"No Wolf" #Outcome:Everyone is fine.

Predicted Label

Another example of MEDICAL TEST: -

- **True Positives (TP): These are cases in which we predicted yes (they have the disease), and they do have the disease.**
- **True Negatives (TN): We predicted no, and they don't have the disease.**
- **False Negatives (FN): We predicted no, but they actually do have the disease. (Also known as a "Type II error.")**
- **False Positives (FP): We predicted yes, but they don't actually have the disease. (Also known as a "Type I error.")**

Accuracy.

It is defined as the ratio of correct predictions to total predictions.

It is calculated as

$$Accuracy = \frac{Total\ correct\ prediction}{Total\ prediction}$$

$$Accuracy = \frac{TN + TP}{TP + TN + FP + FN}$$

Accuracy: Overall, how often is the classifier correct?

Misclassification

It is defined as the ratio of incorrect predictions to total predictions.

$$Misclassification = 1\text{-}Accuracy$$

$$Misclassification = \frac{Incorrect\ prediction}{Total\ prediction}$$

$$Misclassification = \frac{FP + FN}{TP + TN + FP + FN}$$

Misclassification Rate: Overall, how often is it wrong?

Also known as Error Rate.

True Positive Rate: When it's actually yes, how often does it predict yes? Also known as Sensitivity or Recall.

If Recall is 75% that mean it is truly predicted 75% but false prediction is 25%.

True Positive Rate (TPR) =RECALL=SENSITIVITY

It is defined as the ratio of TP to total actual positives.

RECALL=

$$TPR = \frac{TP}{Total\ number\ of\ actual\ positive}$$

$$TPR = \frac{TP}{TP + FN}$$

False Positive Rate (FPR)

It is the ratio of FP to Total Actual Negatives

$$FPR = \frac{FP}{Total\ number\ of\ actual\ negative}$$

$$FPR = \frac{FP}{FP + TN}$$

Email Spam detection: This is one of the example where Precision is more important than Recall.

Precision: This tells when you predict something positive, how many times they were actually positive.

False Positive Rate: When it's actually no, how often does it predict yes?

True Negative Rate (TNR)

It is defined as the ratio of TN to Total Actual Negatives

$$\textbf{TNR} = \frac{TN}{Total\ number\ of\ actual\ negative}$$

$$\textbf{TNR} = \frac{TN}{FP + TN}$$

True Negative Rate: When it's actually no, how often does it predict no? also known as specificity.

Precision

It is the value that has been predicted as positive and how often it is correct

- We use precision when wrong results could lead to loss of business
- It is more useful when FP is a higher concern than FN
- Example:- Email spam(If an important mail is sent to the spam folder and the user misses it due to this reason, It might cause a huge loss to the company, Hence Precision is more useful in this case)

$$precision = \frac{TP}{Total\ predicted\ as\ positive}$$

$$precision = \frac{TP}{FP + TP}$$

Precision: When it predicts yes, how often is it correct?

FOR EXAMPLE: - measure of patients that we correctly identify having a heart disease out of all the patients actually having it.

Recall: - True Positive Rate TPR

Take a big fishing net and throw it into the ocean. You pull it back and what do you have? You've caught some fish but you also caught some garbage old car tires.

Precision = how discerning is your net? Of all the stuff you caught, what % of them are fish?

fish / (# fish + # tires)

Recall = How big is your net? Of all the fish in the ocean, what % did you catch?

fish in your net / # fish in the ocean

F1 Score

- It is the harmonic mean of precision and the recall
- It gives a combined approximation of both precision and recall.
- When False Positive and False Negative both are important parameters for the business F1 score helps.

$$F1\,Score = \frac{2}{\frac{1}{Precision} + \frac{1}{recall}}$$

Support: These values simply tell us how many observations belonged to each class in the dataset.

```
lr_conf_mat=confusion_matrix(Y_test,Y_pred_lr_test)
print(confusion_matrix(Y_test,Y_pred_lr_test))
#Visualization Confusion Matrix
plt.subplots(figsize=(5,5))
sns.heatmap(lr_conf_mat,annot=True,linewidths=0.5,linecolor="red",fmt=".0f",d
ata=X_test)
plt.show()
```

EXAMPLE:-

n=165	Predicted: NO	Predicted: YES	
Actual: NO	TN = 50	FP = 10	60
Actual: YES	FN = 5	TP = 100	105
	55	110	

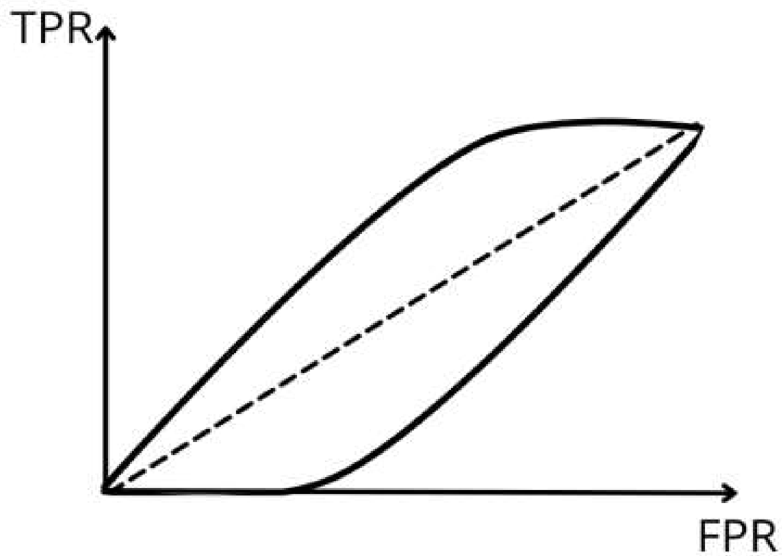
- **Accuracy:** Overall, how often is the classifier correct?
  - (TP+TN)/total = (100+50)/165 = 0.91
- **Misclassification Rate:** Overall, how often is it wrong?
  - (FP+FN)/total = (10+5)/165 = 0.09
  - equivalent to 1 minus Accuracy
  - also known as "Error Rate"
- **True Positive Rate:** When it's actually yes, how often does it predict yes?
  - TP/actual yes = 100/105 = 0.95
  - also known as "Sensitivity" or "Recall"
- **False Positive Rate:** When it's actually no, how often does it predict yes?
  - FP/actual no = 10/60 = 0.17
- **True Negative Rate:** When it's actually no, how often does it predict no?
  - TN/actual no = 50/60 = 0.83
  - equivalent to 1 minus False Positive Rate
  - also known as "Specificity"

- **Precision:** When it predicts yes, how often is it correct?
  - TP/predicted yes = 100/110 = 0.91
- **Prevalence:** How often does the yes condition actually occur in our sample?
  - actual yes/total = 105/165 = 0.64

ROC AUC

ROC AUC Curve

- ROC means Receiver Operating Characteristics.
- This was initially used by operators of the military radar in 1941, World war II
- AUC taken from Economics world.



ROC curve is a graph plotted between TPR and FPR

ROC (Receiver operating characteristic) curve

Area under the curve

The Area under the Curve (AUC) is a summary of the ROC curve and is a measure of a classifier's ability to discriminate between classes.

The higher the area the better the performance of the model to distinguish between the positive and the negative cases.

The confusion matrix helps us visualize whether the model is "mistaken" in distinguishing between two classes.

Positive or **Negative** are the names of the predicted labels by the ML model. Whenever the prediction is wrong, the first word is **False**, and when the prediction is correct, the first word is **True**.

ROC curve is built upon two metrics that are arrived from the confusion matrix: true positives rate (**TPR**) and false positive rate (**FPR**). **TPR** is the same as recall. It is the ratio of the correctly predicted positive samples divided by all actual positive samples available from the dataset. TPR focuses on the actual positive class:

$$\boxed{TPR = \frac{TP}{TP+FN}}$$

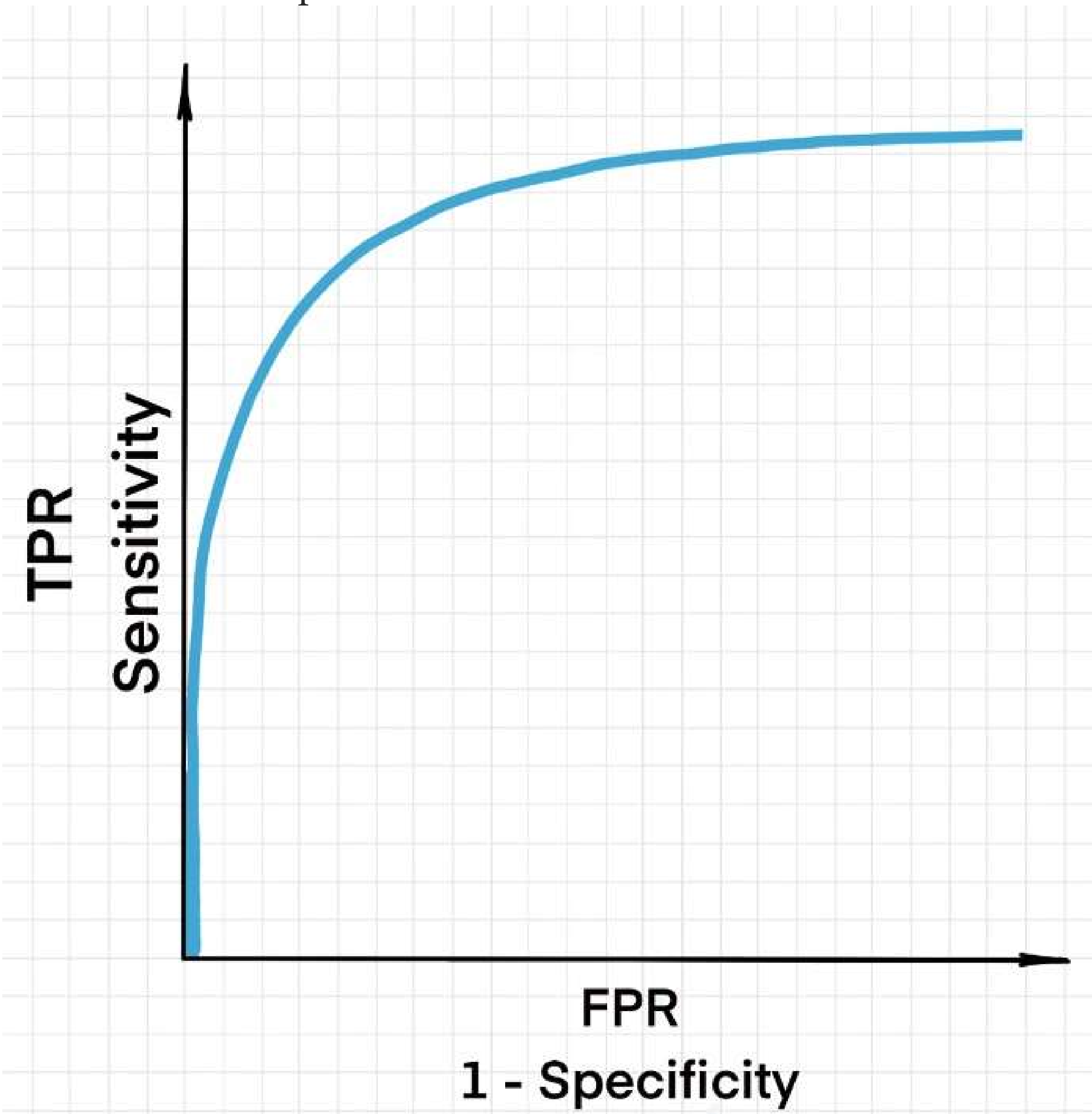
True Positive Rate formula

In turn, FPR is the ratio of false positive predictions to the total number of actual negative samples. It is the same as *1 — Specificity*:

$$\boxed{FPR = \frac{FP}{FP + TN}}$$

FPR formula

The ROC Curve is plotted based on TPR and FPR.



ROC curve example

By using TPR and FPR, the ROC Curve shows your classification model's performance at all classification thresholds.

By default, classification threshold is 0.5. Any probability above 0.5 will be classified as class 1 (positive) and below 0.5 will be classified as class 0 (negative).

So, ROC, in turn, tells us how much the ML model is capable of distinguishing between two classes at various thresholds.